

SIR-Teach: Student-Implicit Reward Teaching

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Abstract

Post-training is critical for turning large language models into effective reasoners, yet current paradigms face limits: reinforcement learning suffers from sparse, high-variance rewards, while naive imitation lacks inductive bias for stepwise reasoning. Teacher-student pipelines partially address this gap, but teachers are often trained with objectives misaligned to their role of producing pedagogical explanations. We present **SIR-Teach** (Student-Implicit-Reward Teaching), a framework that aligns teacher optimization with student learning by conditioning teachers on both question and solution and scoring explanations with a *student-implicit reward* that combines solution likelihood and a KL term encouraging transferable tokens. Optimized with a stable preference-based objective, teachers yield raw traces directly usable for distillation. Experiments show that SIR-Teach outperforms SFT, RL teachers, and curated pipelines across reasoning benchmarks, scales effectively (7B→32B, 1K→17K), provides stronger RL cold-starts, and transfers zero-shot to new domains, highlighting implicit-reward teaching as a principled and practical paradigm for reasoning models.

CCS Concepts

• Computing methodologies → Natural language processing.

Keywords

Reinforcement Learning, Implicit Reward, Knowledge Distillation, Teaching Models

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1 Introduction

Post-training turns general-purpose language models into capable task solvers. Two families dominate today’s practice [15]. The first is *imitation* (supervised fine-tuning, SFT), which learns from demonstrations [14]. The second is *preference-/RL-based* optimization, which learns from evaluative signals and has recently driven progress in long-form reasoning [12]. Despite their success, both paradigms exhibit structural limitations for reasoning models [20]: correctness rewards are sparse and hard to explore from, while

naïve imitation often lacks the inductive bias needed to internalize step-by-step reasoning [6, 18].

A widely adopted remedy is the teacher-student pipeline: a strong model produces “reasoning traces” which are distilled into a smaller student [9]. Yet the objective used to train today’s teachers (solve from scratch under sparse rewards) is misaligned with their downstream role (produce explanations that help a student recover the solution) [24]. In parallel, recent theory shows that post-training objectives can be viewed through *implicit rewards* that emerge from distribution matching-offering a bridge between SFT and preference learning-but this lens is rarely operationalized for explanation-centric teaching [17, 22]. Existing studies have also highlighted that dense pedagogical signals from teacher models can improve distillation efficiency across model scales and domains, while preference-based objectives such as DPO unify RL with pairwise supervision, revealing the potential of implicit-reward formulations to overcome KL-regularization limits [19, 23].

We present **SIR-Teach**, which aligns teacher optimization with student learning: given a question and its verifiable solution, the teacher generates explanations scored by an *implicit reward* that couples the student’s solution likelihood with a teacher-student KL encouraging transferable intermediate tokens. The teacher is trained with a critic-free, DPO-style preference objective relative to a frozen reference policy, and the resulting traces directly supervise student distillation. Theoretically, this training can be viewed as implicit-reward distribution matching that unifies SFT and preference learning. Empirically, across 7B/32B models and 1K/17K budgets, SIR-Teach improves student performance, strengthens RL cold-starts, and transfers zero-shot to new domains.

2 Methodology

2.1 Framework Overview

We consider a teacher-student setting for tasks with verifiable solutions. Let $\mathcal{D} = \{(q_i, s_i)\}_{i=1}^N$ be question-solution pairs. The *teacher* policy $\pi_T(\cdot | q, s)$ generates a stepwise explanation $e = (e_1, \dots, e_L)$, and the *student* policy $\pi_S(\cdot | q, e)$ learns to predict s from (q, e) . Thus, the teacher is optimized not to solve from scratch, but to produce pedagogical traces for distillation.

Regularized objective. Given an implicit reward $r(q, s, e)$ that evaluates e from the student’s perspective, the teacher is updated via KL-regularized policy improvement:

$$\pi_T^* = \arg \max_{\pi} \mathbb{E}_{e \sim \pi(\cdot | q, s)} [r(q, s, e)] - \beta \text{D}_{\text{KL}}(\pi(\cdot | q, s) \parallel \pi_{\text{ref}}(\cdot | q, s)), \quad (1)$$

where π_{ref} is a frozen reference policy and $\beta > 0$ controls a trust region.

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The stationary solution has the Boltzmann form

$$\pi_T^*(e|q, s) = \frac{\pi_{\text{ref}}(e|q, s) \exp(\beta^{-1}r(q, s, e))}{\sum_{e'} \pi_{\text{ref}}(e'|q, s) \exp(\beta^{-1}r(q, s, e'))}, \quad (2)$$

which exponentially favors higher-reward explanations while remaining close to π_{ref} .

Practical update. For each (q, s) , we draw multiple candidates $e_{1:G} \sim \pi_T(\cdot|q, s)$, score them with $r_i = r(q, s, e_i)$, and form a variance-reduced training signal. The teacher parameters θ are updated by

$$J(\theta) = \mathbb{E}_{(q,s) \sim \mathcal{D}, e_{1:G} \sim \pi_T} \left[\frac{1}{G} \sum_{i=1}^G (r_i - \bar{r}) - \beta \text{D}_{\text{KL}}(\pi_T \| \pi_{\text{ref}}) \right], \quad (3)$$

where $\bar{r}$ denotes the group mean. This increases the likelihood of high-reward explanations while regularizing against π_{ref} .

Finally, the optimized traces (q, s, e) directly supervise the student by maximizing $\pi_S(s | q, e)$.

2.2 Implicit Reward Design

We define an *implicit reward* that evaluates teacher explanations from the student's perspective, providing a dense signal that balances solution correctness and explanation transferability.

Student correctness. Given (q, s) and a teacher explanation e , the student policy π_S predicts the sequence $s = s_{1:T}$ under (q, e) with likelihood $\pi_S(s | q, e) = \prod_{t=1}^T \pi_S(s_t | q, e, s_{<t})$, yielding the dense term $r^{\text{SS}}(q, s, e) = \log \pi_S(s | q, e)$.

Explanation transferability. To discourage leakage-like shortcuts, we compare the teacher's explanation token distribution with the student's prior predictive distribution over explanation tokens t_e : $r^{\text{KL}}(q, s, e) = -\text{D}_{\text{KL}}(\pi_T(t_e | q, s) \| \pi_S(t_e | q))$. This rewards explanations whose token patterns are plausible under the student's prior and penalizes over-reliance on s . By Pinsker's inequality [4], $|\Pr_{\pi_T}(A) - \Pr_{\pi_S}(A)| \leq \sqrt{\frac{1}{2} \text{D}_{\text{KL}}(\pi_T \| \pi_S)}$ for any event A , so controlling the KL uniformly bounds deviations in explanation patterns, including explicit answer copying.

Combined implicit reward. We combine the two components as

$$r(q, s, e) = r^{\text{SS}}(q, s, e) + \lambda r^{\text{KL}}(q, s, e), \quad \lambda \geq 0, \quad (4)$$

balancing accuracy and transferability. Owing to the KL-regularized update in (1)–(2), the objective is affine-invariant in r : replacing r by $ar + b$ with $a > 0$ leaves π_T^* unchanged up to $\beta \mapsto \beta/a$.

Unified view via f -divergence. For distributions p and q over explanations e , the f -divergence admits the Fenchel–Legendre dual

$$D_f(p \| q) = \sup_{u: \mathcal{E} \rightarrow \mathbb{R}} (\mathbb{E}_{e \sim p}[u(e)] - \mathbb{E}_{e \sim q}[f^*(u(e))]). \quad (5)$$

Choosing $p = \pi_T(\cdot | q, s)$ and $q = \pi_S(\cdot | q)$ shows that maximizing a suitable implicit score $u(e)$ under the teacher while regularizing it under the student recovers an f -divergence objective. In SIR-Teach, $u(e)$ is instantiated by the student-likelihood signal r^{SS} , while the KL term r^{KL} plays the role of regularization; hence (4) is a tractable surrogate with a clear pedagogical interpretation.

2.3 Optimization with Preference-based Learning

Directly applying reinforcement learning with scalar advantages to $r(q, s, e)$ can be high-variance and unstable. Instead, we train from

pairwise preferences over explanations: for each (q, s) , construct a pair (e^+, e^-) with $r(q, s, e^+) > r(q, s, e^-)$ and fit the teacher to prefer e^+ over e^- relative to a frozen reference policy. This follows Direct Preference Optimization (DPO) [19] and avoids explicit reward models or critic networks.

Objective. The teacher is optimized with the DPO-style loss

$$\mathcal{L}_{\text{pref}}(\theta) = -\mathbb{E}_{(q,s,e^+,e^-)} \left[\log \sigma \left(\beta \log \frac{\pi_T(e^+|q,s)}{\pi_{\text{ref}}(e^+|q,s)} - \beta \log \frac{\pi_T(e^-|q,s)}{\pi_{\text{ref}}(e^-|q,s)} \right) \right], \quad (6)$$

where π_T is the teacher, π_{ref} the frozen reference, $\sigma(\cdot)$ the logistic function, and β a scale. This encourages higher-implicit-reward explanations while keeping the policy close to the base model.

Interpretation. The loss admits an f -divergence view: it aligns the teacher with an implicit-reward-weighted target while regularizing toward the reference; choosing total variation recovers standard SFT, placing SFT and preference optimization on the same implicit-reward spectrum. From a logit perspective, language model logits approximate token-level Q -values, and preference learning shapes *relative* differences between explanations rather than regressing to absolute returns-yielding smoother gradients and more stable convergence.

In sum, combining implicit rewards with a DPO-style objective provides a stable and theory-backed procedure for training teachers to produce pedagogical, distributionally aligned explanations.

3 Experiments

3.1 Experimental Setup

Models and Backbones. We report results on two student scales: a 7B model and a 32B model. Our main experiments use the Qwen2.5 family [1], and we also include results on a Mistral-7B variant [13] for diversity of backbones. Unless noted, the *teacher* is a 7B model initialized from the same family as the student. The reference policy π_{ref} is the frozen base checkpoint matching each teacher.

Data and Tasks. Training uses a verifiable reasoning corpus of $\sim 17\text{K}$ math/coding question-solution pairs; we also evaluate a 1K randomly-sampled subset to simulate low-budget settings. For evaluation we follow common reasoning benchmarks: *AIME24* [10], *MATH-500* [11], and *GPQA-Diamond* [2]. To assess out-of-domain transfer, we include the *Countdown* [7] arithmetic composition task. Only the question q and ground-truth solution s are used during training; labels are verifiable and permit exact-match scoring.

Teacher Training (Preference-Based). We train the teacher with a DPO-style objective on explanation pairs ranked by the implicit reward $r(q, s, e)$. For each (q, s) we sample multiple candidates, form (e^+, e^-) by reward ordering, and optimize against a frozen reference policy to stay close to the base model. A brief SFT warm-up is used for format adaptation [25].

Student Distillation. Each (q, s, e) from the teacher is reformatted into a student prompt, and the student is fine-tuned to maximize $\pi_S(s | q, e)$. Distillation is run with 1K and 17K budgets, without external verifiers or post-processing. The reward trade-off λ is tuned, with default 0.3.

Baselines. (1) *SFT-only* [5] and *SFT→DPO/SimPO* [16]. (2) *RL-trace pipelines* (e.g., s1/Sky-T1/Bespoke). (3) *RLT traces* [3]. (4) *Direct RL*

(GRPO) [21] and *cold-start RL* with each trace source. All baselines share the same student backbones and 1K/17K budgets.

Evaluation Protocol. We report pass@1 accuracy using LightEval-style execution [8] with identical decoding across methods. Results are averaged over three seeds with 95% CIs. For transfer, models are trained in-domain and evaluated zero-shot on *Countdown*.

3.2 Main Results: Student Performance

Table 1 compares *Qwen* and *Mistral* backbones, showing that **SIR-Teach** yields the strongest students under both 1K and 17K budgets. Gains are most pronounced at 1K and remain when scaling to 32B. This suggests student-centered implicit rewards generate explanations that distill more effectively than large closed-source teachers or post-processed pipelines.

Table 1: Main results across Qwen (7B/32B students) and Mistral (7B students) families under different data budgets. Reference models are shown only for context.

Model / Method	Data	AIME24	MATH500	Overall
<i>Reference (not directly comparable)</i>				
QwQ-32B	N.A.	50.0	90.6	65.0
DeepSeek-R1	800K+	79.8	97.3	82.9
<i>Qwen - 7B student, 1K distillation samples</i>				
SFT-only (small-LR)	1K	12.0±0.8	76.0±0.8	41.4±0.8
SFT → DPO	1K	17.0±0.8	79.0±0.7	45.2±0.7
SFT → SimPO	1K	18.2±0.8	79.8±0.7	46.2±0.7
s1-7B (pipeline)	1K	13.3±0.9	80.0±0.8	42.4±0.8
Bespoke-7B (pipeline)	1K	16.7±0.8	80.2±0.7	45.0±0.7
SIR-Teach-7B (ours)	1K	22.1±0.7	82.1±0.7	49.0±0.7
<i>Qwen - 7B student, 17K distillation samples</i>				
SFT-only (small-LR)	17K	18.1±0.5	81.0±0.5	45.3±0.5
Bespoke-7B (pipeline)	17K	20.0±0.5	82.0±0.4	46.6±0.5
SIR-Teach-7B (ours)	17K	24.1±0.4	83.2±0.4	50.2±0.4
<i>Qwen - 32B student, 1K distillation samples</i>				
s1-32B (pipeline)	1K	50.0±0.5	92.6±0.4	66.4±0.5
s1-32B + budget	1K	56.7±0.5	93.0±0.4	69.8±0.5
Bespoke-32B (pipeline)	1K	46.7±0.6	92.6±0.5	65.6±0.5
SFT → DPO	1K	57.3±0.4	92.4±0.5	69.4±0.4
SIR-Teach-32B (ours)	1K	61.2±0.4	94.3±0.3	71.8±0.4
<i>Qwen - 32B student, 17K distillation samples</i>				
Sky-T1-32B (pipeline)	17K	43.3±0.4	82.4±0.3	60.8±0.4
Bespoke-32B (pipeline)	17K	63.3±0.3	93.0±0.3	71.5±0.3
SIR-Teach-32B (ours)	17K	67.8±0.3	93.6±0.3	73.9±0.3
<i>Mistral - 7B student, 1K distillation samples</i>				
SFT-only (small-LR)	1K	10.8±0.9	74.5±0.8	39.8±0.8
SFT → DPO	1K	15.6±0.8	77.4±0.7	43.5±0.8
SFT → SimPO	1K	16.3±0.8	78.1±0.7	44.3±0.8
s1-7B (pipeline)	1K	12.5±0.9	78.9±0.8	41.5±0.9
Bespoke-7B (pipeline)	1K	15.9±0.8	79.1±0.7	44.0±0.8
SIR-Teach-Mistral-7B (ours)	1K	20.3±0.7	81.0±0.7	47.5±0.7
<i>Mistral - 7B student, 17K distillation samples</i>				
SFT-only (small-LR)	17K	16.2±0.5	80.2±0.5	44.0±0.5
Bespoke-7B (pipeline)	17K	18.9±0.5	81.5±0.4	45.6±0.5
SIR-Teach-Mistral-7B (ours)	17K	22.7±0.4	82.6±0.4	49.1±0.4

3.3 Cold-Start RL with SIR-Teach

We evaluate SIR-Teach as a *cold-start* for RL: starting from Qwen-7B, we run GRPO with correctness rewards under four initializations-(i) none, (ii) raw R1 traces, (iii) curated pipelines (s1/Bespoke), and (iv) SIR-Teach explanations selected by implicit-reward top- k ($k=1$).

As shown in Table 2, SIR-Teach attains the best overall performance, including +13.2 on AIME24, and it improves MATH500 and overall accuracy over curated pipelines without post-processing. Ablations show limited sensitivity to larger k , sharp drops without reward-based ranking, and strong results even with smaller teachers-supporting SIR-Teach as a simple, robust cold-start for RL.

Table 2: Cold-start RL on Qwen-7B.

Initialization	AIME24	MATH500	GPQA-D	Overall
RL (no cold-start)	13.2±1.0	74.4±0.9	34.8±1.2	40.8±1.0
RL + R1-raw traces	11.1±0.9	71.2±0.8	34.9±1.0	38.7±1.0
RL + Pipeline traces	16.8±0.7	78.3±0.7	36.9±0.8	44.0±0.7
Bespoke-7B + RL pipeline	16.9±0.5	82.7±0.6	45.4±0.7	48.2±0.7
RL + SIR-Teach (ours)	26.4±0.4	84.1±0.5	43.0±0.6	50.6±0.5

3.4 Out-of-Domain Transfer

We evaluate cross-domain generalization on the *Countdown* arithmetic composition task, using three settings: *Zero-shot*, *Few-shot* (32), and *Few-shot* (128). For each setting, we compare *Direct RL*, *Pipeline traces*, and *SIR-Teach* by training students under identical protocols and reporting pass@1. Figure 1 shows that SIR-Teach consistently dominates across all regimes, with the largest margins in the zero-shot case, suggesting that student-centered implicit rewards yield explanations with stronger cross-task transferability.

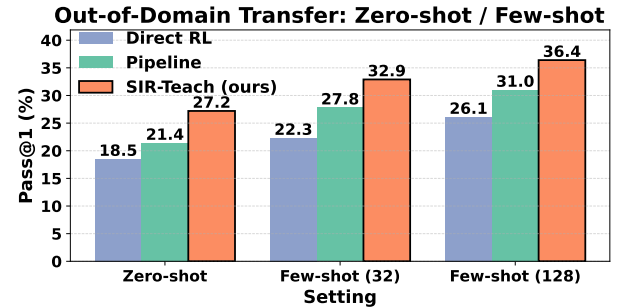


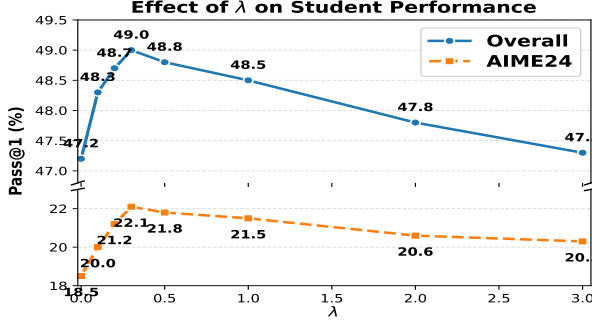
Figure 1: Out-of-domain transfer on Countdown. Bars show pass@1 (%) under Zero-shot, Few-shot (32), and (128) settings.

3.5 Ablations on the Implicit Reward

We analyze the contribution of reward components, the role of the trade-off λ , and the agreement between reward ranks and downstream learning, with a note on leakage. Table 3 compares using only r^{SS} , only r^{KL} , and the combination on Qwen-7B under 1K/17K budgets. The transferability term alone is weaker, but its combination with the student-likelihood term consistently yields the best accuracy. Figure 2 then plots Overall and AIME24 versus $\lambda \in \{0, 0.1, 0.3, 1.0, 3.0\}$, showing a clear peak around $\lambda=0.3$, which balances correctness and transferability. Correlation analysis further confirms that reward rankings align with downstream gains: Kendall's $\tau = \{0.66, 0.61, 0.58\}$ and Spearman's $\rho = \{0.84, 0.79, 0.76\}$ on AIME24/MATH500/GPQA-D ($p < 10^{-4}$). Finally, leakage is reduced when the KL term is weighted more: on Qwen-7B (1K), the rate drops from 7.8% at $\lambda=0$ to 5.0% at $\lambda=0.3$.

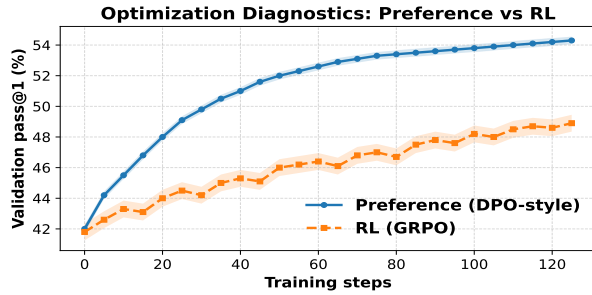
Table 3: Component ablation on Qwen-7B.

Reward used	1K		17K	
	AIME24	Overall	AIME24	Overall
Only r^{SS}	20.1 $\pm$ 0.5	47.2 $\pm$ 0.6	22.6 $\pm$ 0.4	48.9 $\pm$ 0.4
Only r^{KL}	14.4 $\pm$ 0.6	43.1 $\pm$ 0.5	17.3 $\pm$ 0.6	44.7 $\pm$ 0.5
$r^{SS} + \lambda r^{KL}$	22.1 $\pm$ 0.2	49.0 $\pm$ 0.4	24.1 $\pm$ 0.4	50.2 $\pm$ 0.3

**Figure 2: Effect of λ on Qwen-7B (1K).**

3.6 Optimization Diagnostics: Preference vs RL

To better understand why preference-based optimization is stable, we compare its learning dynamics with GRPO reinforcement learning. Figure 3 plots validation pass@1 over training steps. Preference learning improves steadily with low variance, while GRPO exhibits higher fluctuations and slower convergence. This supports the view that preference objectives optimize relative differences between explanations, providing smoother gradients than scalar RL rewards. Consistent with our analysis that preference logits approximate Q -values, this stability highlights the advantage of preference supervision in reasoning tasks.

**Figure 3: Training dynamics: preference optimization vs. GRPO—faster and more stable convergence.**

4 Conclusion

We presented **SIR-Teach**, a teacher-student framework where explanations are evaluated from the student’s perspective via an implicit reward combining solution likelihood and transferability. This yields pedagogical traces that scale across model sizes and data budgets. Experiments show consistent gains over SFT, preference optimization, and pipeline distillation, while also enabling effective RL cold-starts and robust out-of-domain transfer. These results

highlight implicit-reward teaching as a practical and theoretically grounded approach for aligning reasoning models.

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